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⇒ Python Syllabus

1. Introduction of python
2. Commands
3. Keywords
4. data types
5. operators
6. input / output statement
7. conditional statement
   if , if else , continuous if else / if else ladder , nested if else
8. loops
   for loop , while loop , nested loop
9. Built in data structures
   data list , tuple , set , dictinary
10. functions
    - Built in functions
    - user defined functions
    - lambda functions
11. file handling
12. error and exception handling

⇒ python libraries for Data Analytics :

- pandas - data manipulation
  1) Insert , update , delete
- numpy - Numerical python for all mathematical and higher dimension data calculation
- Matplot lib's pyplot
- Sea born

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⇒ python libraries for machine learning

* sklearn (Scikit-learn) for all kind of machine learning tasks

⇒ python libraries for deep learning
* tensorflow with keras
* pytorch

- Extension of python file = .py
- extension of ~~python~~ Jupyter notebook file = .ipynb

⇒ Kernal : Engine that runs the python code

1) code - used to write and execute the python code

2) Markdown - It can be used for heading or content for the purpose so cell number won't come. And the output comes in blank space [ ]

3) raw - Whatever we write it will be inside the cell itself in the same format

We use .format() and f string methods to insert the variable values directly inside a string. It makes code readable, avoids type conversion errors and allows formatting numbers easily

⇒ Example → variable initialization

① name = "Chandana" → value
Age = 21
print ("My name is name and age is age")
(variable name, box)

* In this case it will print as it is inside the print brackets
o/p → My name is name and age is age.

* But if we want to print the actual value of variable need to write outside of "" and with comma

print ("My name is", name, "and my age is", age)

o/p : My name is Chandana and my age is 21

1) .format() method - {} - placeholder
. Then .format (variable) fills the place holder

name = Chandana
age = 21

print ("My name is {} and my age is {}".format(name, age))

2) f - string method
we add f before string
write variable directly inside {} placeholder

print(f"My name is {name} and my age is {age}")

O/P My name is chandana and my age is 21

⇒ What is programming?

- It's sets of instruction given to a machine to perform a specific task

⇒ Three levels of programming

1) Low/binary level programming
2) Assembly/Mid level programming
3) High level programming

⇒ Python

Python is a high level, interpreted object-oriented, indented programming language

* Complier - A complier translates/process the entire source code into machine code before execution if errors are found, the code must be corrected & re complied takes extra time, for large programs

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* Interpreter - An interpreter executes a program step by step line by line and stops when it encounters an error

⇒ Features

* It's easy to learn because of its simple syntax
* It's open source → free
* Platform independent
  - Scripts windows, Linux and Mac
* Huge library, community support
* Used of almost all kind of software development

⇒ Applications

* Web applications
* desktop application
* Gaming application
* Mobile Application
* data analytics and data science
* AI
* automation
* IoT
* graphics
* data security

The developer of python is guido van Rossum

python 1 : 1991
python 2 : 2000
python 3 : 2008

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Assignment - 01

1) Introduction of programming and python programming
2) History of python
3) Advantages and disadvantages of python

2) History of python

* python was created by Guido van Roussum
* Its development began in the late 1980's
* python was first released in 1991
* Its name was inspired by Monty python's flying circus
* python 2.0 was released in 2000
* python 3.0 was released in 2008

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=> Advantages

1) easy to learn and use
2) fast and open-source
3) simple and readable syntax
4) supports many libraries
5) used in data analytics, AI and automation

=> Disadvantages

1) slower than some complied languages
2) can use more memory
3) less commonly used for native mobile apps
4) some errors appear only during execution

Types of operators

a = 5
b = 3

5 + 3 = 8 Add
5 - 3 = 2 Sub
5 * 3 = 15 Multi
5 / 3 = 1.5 -> 1.5 / 3

3) 5 (1 9 Modulus → Remainder 5 % 3 = 2
3 / 2 → R

5 // 3 = 1 → Quotient
5 ** 3 = 125

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* Comments :-

They are the part of programming but they are not part of execution because they used to understand the flow programming, sometimes for understanding the logics behind the code

Types of comments

1) Single line comment
2) multi line comment / docstring method

Ex:- # my first python program

''' program '''
""" program """

* keywords :
They are reserved words used to perform some specific tasks.
ex:- break, if, else for

keywords are not used as identifiers (variables, function names)

import keyword
print(keyword.kwlist)
It gives all the keywords present in python

* Variables

They are the identifiers used to store value to them.

Valid variable declaration

a = 34
A = 12
Num = 56
Num1 = 76
stu_name = "Chandana"
stu_id = 1764

Invalid variable declaration
emp name = 'abc'
emp@place = 'bengaluru'
1Num = 45
2Num = 96

* Data Types :-

It will tell about the type of data we are having

① Single valued data type
It can store only value in variable
ex: int, float, complex, boolean and etc

② Multi valued data type
It can store multiple values in a variable
ex: list, tuple, dictionary, set

* Python operators :-

It is the symbol which performs certain operation between two operands

S = a + b
a, b, S → operands
=, + → operators

⇒ Types of operators

① Arithmatic operators
+ = - / % // **

② Relational operators
== != > < >= <=

③ assignment operators
+= -= *= /= //= %= **=

④ logical operators
and or not

* And = Any one condition should be true then it returns true

* OR = When Both the condition is true then it returns "True"

* NOT = It results opposite result if it is true it tells false & vice versa

⑤ Membership operators
in not in

⑥ Bitwise operators
& | ~ >> <<

27/10/2021 Conditional statements

⇒ Conditional statement are used when the statements have to print based on condition.
or
when we use conditional statements when we have multiple different statements to print

① # if else Syntax

```
age = int(input("enter your age:"))
if (age <= 18):
    print("your eligible")
else
    print("your not eligible")
```

② # if Syntax

```
age = int(input("enter your age:"))
if (age <= 18):
    print("your eligible")
```

③ # continuous if else Syntax

```
if (cond 1):
    statement 1
elif (cond 2):
    statement 2
elif (cond 3):
    statement 3
else:
    default statement
```

⇒ Loops

A loop is a programming concept used to execute a block of code repeatedly until certain condition is met of the sequence is completed

* Types of loops in python

① For loop
   a) for loop with sequence
   b) for loop with range
② Nested loop
   a) for loop inside for loop
   b) for loop inside while loop
   c) while loop inside while loop
   d) while loop inside for loop

① For loop with sequence syntax
iterative means num will be changing

for iterative variable in sequence variable:
    statements

Ex: name = "chandana Yadav"
for d in name:
    print(d)

o/p
c
h
a
n
d
a
n
a
Y
a
d
a
v

② # for loop with range syntax
initialization - start value
condition - stop value
incr/decr - step size

range(start value, stop value, step size)

It takes only numerical values can't pass float or alphabet

default start value 0
stop value is upto given condition
step size 1

For iterative variable i in range (strV, stp, stop):
    Var = , Var < , V+1
    statements

Ex: for i in range (1, 6, 1):
    print("i = " i)
    print("hello")

o/p
i = 1
hello
i = 2
hello
i = 3
hello
i = 4
hello
i = 5
hello

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(3) while loop syntax

without inc/dec do not execute this while loop it goes to infinity

initalization
while (condition):
    statements
    inc/dec

Ex:
```
i = 1
while (i<=10):
    print(i)
    i = i+1
```

o/p
1
2
3
4
5
6
7
8
9
10

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